Mainstreaming the tricot approach into the CGIAR-NARES Breeding Initiative

Insights from the meeting held in Nairobi, 3-5 December 2024

CIMMYT
International Maize and Wheat Improvement Center

Alliance
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INTRODUCTION

A meeting took place at CIMMYT campus in Nairobi, from December 3-5, 2024, bringing together project partners, researchers, and stakeholders to discuss progress, challenges, and future directions for on-farm testing using the triadic comparison of technology options (tricot) approach[1]. The meeting was convened by the Dryland Crops Program (DCP) and 1000FARMS Platform, two complementary initiatives that aim to enhance breeding and seed system efficiencies across Africa. The goal was to provide a platform for cross-learning and collaboration, aligning efforts between the two programs and the CGIAR Initiatives on Market Intelligence, RTB Breeding, Seed Equal, and CIMMYT's Maize Research Team.

The Dryland Crops Program (DCP), implementing the Accelerated Varietal Improvement and Seed Systems in Africa (AVISA) project, is dedicated to improving the livelihoods of small-scale producers and consumers of sorghum, millet, groundnut, cowpea, common bean, finger millet, pigeon pea, and chickpea. Despite growing interest in dryland crops due to their climate resilience, the adoption of improved seed varieties remains limited due to information gaps, seed system inefficiencies, and limited access to early-generation seed. To address these challenges, DCP focuses on establishing synergistic networks that connect researchers, extension services, and seed producers, supporting national agricultural research systems (NARES) with research and professional development. DCP integrates the CGIAR Excellence in Breeding (EiB) platform to improve crop improvement networks, leveraging CGIAR and NARES collaboration to optimize breeding pipelines.

The 1000FARMS Platform aims to establish an Africa-wide network for on-farm testing, addressing a critical gap in on-farm data collection for breeding programs. Breeding programs in Africa lack substantial on-farm data to evaluate the performance of new varieties in real-world conditions. 1000FARMS closes this gap by leveraging digital technologies, citizen science, and data science to scale up on-farm data collection, creating a network of digital trial farm sites to monitor production, farmer preferences, and climate conditions, and enabling late-stage selection by generating accurate data on new variety performance to ensure they are farmer-preferred, climate-adapted, and scalable. As a CGIAR-NARES platform, 1000FARMS enhances the impact of breeding networks by supporting on-farm testing at scale, providing data-driven insights that improve variety selection and deployment.

[1] de Sousa et al. (2024) Agron. Sustain. Dev. 44, 8 (2024). https://doi.org/10.1007/s13593-023-00937-1

MEETING **OVERVIEW**

A total of 26 participants attended the in-person meeting which included funders, partners from CGIAR and NARES. The first day of the meeting focused on understanding who participates in tricot and where on-farm verification trials (OFVT) are taking place. After an introduction by the hosts, representatives from Gates Foundation shared their perspectives. Presentations on the current scale of on-farm trials using the tricot method followed, highlighting crop diversity, geographic spread, and socioeconomic profiles of participating farmers. The morning also featured discussions around sampling strategies, agroecological representativeness, and trial site selection.

OBJECTIVES

a) **Understand how tricot trials are conducted and share experiences among the CGIAR-NARES network.**
b) **Foster cross-learning and coordination among crops, countries, and institutions.**
c) **Co-develop strategic frameworks and tools for scaling.**
d) **Identify mechanisms to strengthen the implementation of tricot for On-Farm Variety Trials (OFVT) across the CGIAR-NARES.**

Breakout sessions engaged participants in reflecting on how to improve farmer sampling, genotype selection, and ClimMob functionality. In the afternoon, presentations delved into how tricot trials are conducted, including standard operating procedures (SOPs), farmer-led yield assessments, and the role of tricot to guide seed systems investments. Another breakout focused on selecting materials for testing and defining protocols for reliable, scalable data collection. The day ended with plenary presentations and discussions, developing operational ideas and emerging questions.

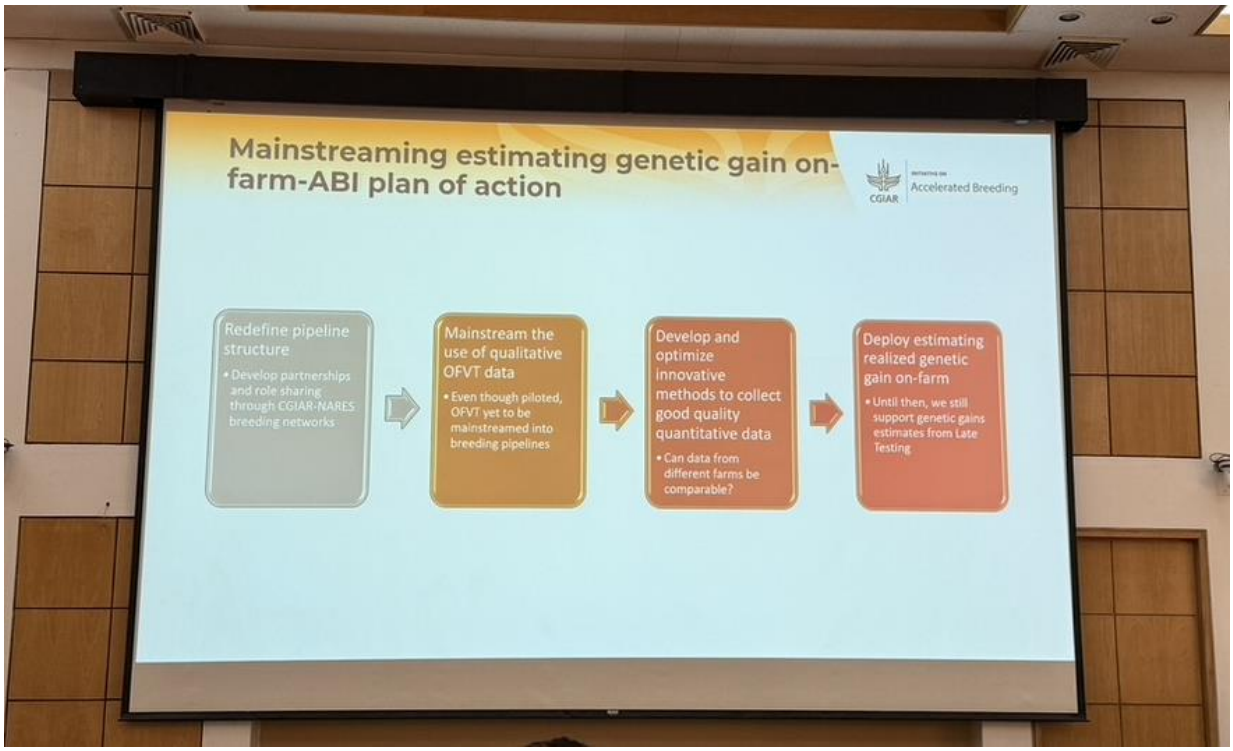
The second day focused on what happens with tricot data and how to strengthen its utility in breeding and seed systems. The morning began with reflections on the previous day and moved into presentations showcasing how tricot data is used for varietal advancement in different crops and countries, Topics included data analysis practices, varietal release processes, and targeting strategies. Key innovations were presented, such as a pipeline for estimating on-farm genetic gains and integrating tricot into product design. A breakout session focused on aligning traits collected with target product profiles (TPPs), improving data reporting, and enhancing trial design.

In the afternoon, attention shifted to operational aspects of tricot: challenges in implementing trials, cross-crop coordination, and seed system bottlenecks. Another breakout session addressed cost control, cross-crop learning, and implementation bottlenecks, leading into a plenary that consolidated lessons and practical solutions. The day ended with a social dinner to foster informal connections among participants.

The third and final day centered on strategic planning and future directions. It opened with a synthesis of key insights from the first two days, followed by a brainstorming session on the barriers to adopt tricot for OFVT. The morning session concluded with a breakout session on

the four cross-cutting themes identified in the meeting: 1) Advancement processes to make decisions on OFVT; 2) Cross-crop coordination network; 3) Digital ecosystem; 4) Standard Operating Procedures (SOPs). Breakout discussions allowed participants to develop actionable recommendations for each theme.

In the afternoon, each group presented its recommendations, which were refined through open feedback. A roadmap session followed, where responsibilities, timelines, and milestones were defined for follow-up actions. The meeting concluded with closing remarks from project leads and BMGF representatives, who highlighted key outcomes and reinforced the commitment to collaborative action for the success of 1000FARMS and tricot mainstreaming. Next sessions in the report summarize the discussions in the breakout sessions and bring key takeaways from these discussions. We close the report with a summary of key priorities and tasks for the next 12 months identified by the four cross-cutting themes groups.



Sampling and characterization

The first breakout session of the meeting focused on one of the foundational elements of tricot implementation: “how to ensure that sampling of farmers and trial sites is both representative and strategic?”. Participants were sub-divided into three groups to discuss the pillars of geographic and demographic representativeness, offering nuanced perspectives on how tricot trials can be strengthened through robust sampling techniques.

Participants agreed that representativeness begins with a solid understanding of where crops are grown and who grows them. They emphasized the importance of mapping production areas and defining Target Populations of Environments (TPEs) to guide the spatial dimension of trial site selection. While geospatial data on crop areas is increasingly accessible, demographic data remains less so. The groups recommended using existing datasets, such as LSMS-ISA surveys and national census data, to better understand farmer typologies (considering factors such as gender, household structure, land size, poverty level, and market orientation). Local extension services were seen as a valuable source of contextual knowledge, though participants also cautioned against allowing these actors to dominate or bias the selection process.

Inclusivity was a recurrent theme, with strong calls to ensure that women, youth, smallholders, and progressive farmers are meaningfully included in tricot trials. To do this effectively, stratified and cluster sampling techniques were seen as essential tools. Some groups proposed using snowballing methods within clusters or farmer organizations to diversify sampling, while also recognizing that expanding sample sizes to capture all demographics could raise costs significantly. Thus, the importance of balancing representativeness with logistical feasibility was acknowledged.

Farmer demographics were not seen as just background variables, but as central to decision-making throughout the trial cycle. Participants pointed out that different demographics bring varying levels of resource access and crop management capacity, which directly affect trial outcomes. Disaggregating data by demographic variables (such as management level or yield category) can improve the robustness of conclusions and support more precise targeting in breeding strategies. In this way, demographic insights become critical inputs to breeding pipelines and the design of production packages tailored to specific market segments.

The question of whether farmer and site selection should be standardized or crop-specific sparked discussions. While many saw value in developing general guidelines based on TPEs and market segments, they also acknowledged the need for flexibility in crops with distinct traits or management demands. For example, while beans and maize may be grown in overlapping agro-ecologies, their breeding priorities and farmer networks may differ, warranting tailored sampling approaches. The groups proposed developing standard tools and frameworks for selection, but with enough room for adaptation to the realities of each crop and country.

In terms of practical recommendations, the groups emphasized the value of re-engaging farmers who had previously provided high-quality data, using past performance to inform future participation. Trial sites should be selected to allow analysis in blocks or homogeneous units, facilitating both statistical robustness and operational feasibility. Participants also encouraged the strategic spacing of sites to ensure coverage and overlap across seasons and geographies, and called for guidelines to avoid biased selection, especially when relying on local intermediaries.

The discussions also highlighted several ways to improve the ClimMob platform (https://climmob.net), the digital backbone of tricot trials. Participants felt that ClimMob should support more sophisticated sampling strategies, incorporating agro-ecological and demographic layers into its design phase. Tools that allow users to visualize historical trial data, map farmer demographics, and check connectivity across trials would significantly strengthen planning and analysis. Moreover, adding dashboards to track emerging trends and enabling binary splits by yield or other key variables would enhance user experience and analytical power.

Finally, the groups reflected on the broader question of stakeholder alignment. Implementing tricot at scale requires more than technical tools and depends on shared commitment across partners. Participants urged a move away from transactional relationships with service providers and toward genuine partnerships built on common goals. Engaging regulators and national systems with a clear value proposition was seen as essential for embedding tricot within variety release processes.

KEY TAKEAWAYS FROM SESSION 1: SAMPLING AND CHARACTERIZATION

1. Identify and visualize total crop production zones at national and sub-national levels to guide site selection based on agro-ecological zones.
2. Leverage existing datasets (e.g., LSMS-ISA, census, RHoMIS, extension surveys) to capture key variables like gender, age, landholding size, and household structure when defining target populations.
3. Use sampling approaches that allow for diversity across agro-ecologies and farmer typologies while ensuring statistical balance and avoiding biases introduced by intermediaries.
4. Avoid over-sampling in the name of inclusivity—optimize for both demographic representation and cost-effectiveness to keep trials manageable.
5. Prioritize farmers with a track record of timely and reliable data submission to build continuity and data quality across seasons.
6. Create general protocols for farmer and site selection based on agro-ecological and market segment frameworks, while allowing for crop-specific adaptations.
7. Incorporate modules in ClimMob for stratified sampling, demographic overlays, mapping of past trials, and dashboards for tracking sampling trends and trial connectivity.
8. Select sites that allow for spatial replication and internal comparison, which can improve the robustness of trial outcomes and reduce variability.
9. Systematically include underrepresented groups—especially women, youth, and smallholders—through intentional recruitment strategies and criteria.
10. Engage national partners, regulators, and farmer organizations early with a clear value proposition for tricot as a reliable tool for variety release and adoption.

Trial design and implementation

The second breakout session of the meeting focused on refining how tricot trials are designed and conducted, with a particular emphasis on improving Standard Operating Procedures (SOPs), aligning trials with breeding and market objectives, and embedding tricot more deeply into national variety release systems.

Participants from all groups agreed that the foundation for high-quality and scalable tricot trials lies in the development and consistent application of clear, well-documented SOPs. These should be purpose-driven and adaptable to varying trial objectives such as variety verification, seed system targeting, or market intelligence. To ensure data quality and relevance, SOPs should include clear guidance on entry selection, trait measurement, and data collection tools, while also being simple enough for field agents and enumerators to apply effectively.

A recurring theme in the discussion was the need for standardized yet flexible entry selection (selected varieties to be tested)**.** Groups agreed that trial entries must be selected based on their intended purpose and aligned with breeding pipelines and Target Product Profiles (TPPs). For example, entries in late-stage trials or OFVTs should meet TPP thresholds and ideally be sourced from the national performance trials (NPTs). While there was some debate over the need to classify tricot trials based on their function (e.g., pipeline vs. released varieties), the consensus was that such distinctions might be necessary in the short term but should ultimately be phased out as tricot becomes mainstreamed as a unified trialing approach.

The classification of tricot trials sparked nuanced discussion. Some participants felt that distinguishing between verification, seed system targeting, and commercial trials could be useful for clarity and methodological rigor. Others emphasized that long-term integration of tricot into national and institutional systems would benefit from a single, flexible framework rather than fragmented protocols. This unified approach would also help promote tricot as a scalable, reliable platform for on-farm testing, without overcomplicating trial design.

Another key point of agreement was the need for adaptable protocols for different use cases, especially when trials are designed to support seed systems. These situations often require varied sample sizes, traits, and levels of data sensitivity. To ensure consistency across these diverse use cases, participants proposed using a set of common check varieties and metadata fields (participant characterization questions and traits to be assessed) that remain stable while allowing for customization based on specific objectives.

The integration of tricot into national variety release systems was recognized as both a challenge and an opportunity. Participants stressed the importance of engaging variety release authorities directly and proactively, using existing meetings and channels to demonstrate the value of tricot-generated data. Many countries already require yield data for variety release, and adapting tricot's yield protocols to meet those requirements could position it as a natural tool for national decision-makers. However, to do this effectively, SOPs must be adapted to reflect the actual data demands of these systems, including simplified yield measurements and harmonized data reporting templates.

Several practical suggestions were raised for improving tricot implementation. These included defining minimum and maximum numbers of entries per trial, providing alerts on the number of sites to sample based on entries, and incorporating tools to factor in attrition and seed loss. Participants recommended standardizing measurement apparatus and plot sizes where

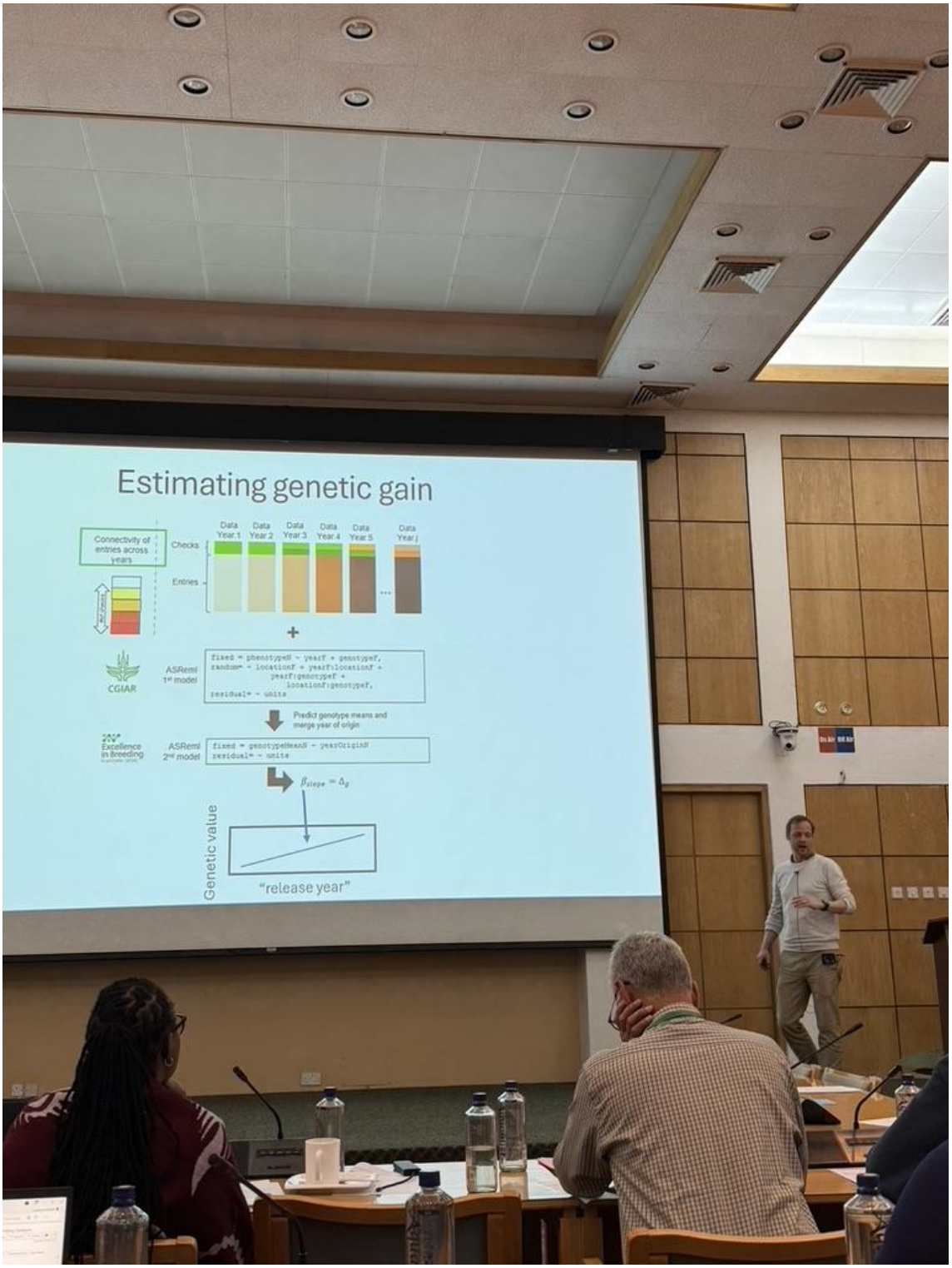
possible and focusing on essential traits aligned with TPPs rather than asking farmers to evaluate unnecessary traits. Pre-season training of stakeholders, especially those collecting data, was seen as crucial for ensuring protocol adherence and data quality.

Importantly, the session emphasized the need to frame tricot not as the end goal, but as a platform that enables scalable, data-driven decision-making. Its power lies in its flexibility and ability to generate insights that support variety adoption, breeding pipeline optimization, and seed system alignment. To that end, participants recommended integrating new analytical tools (such as yield validation and trial classification modules) into ClimMob to further enhance trial management and data interpretation.

In conclusion, the session made a strong case for better harmonization, user-focused training, and data-driven clarity in trial design. These improvements will help embed tricot more deeply into national systems, strengthen its contribution to breeding and scaling, and ultimately support more inclusive and effective agricultural innovation.

KEY TAKEAWAYS FROM SESSION 2: SAMPLING AND CHARACTERIZATION

1. Update SOPs to reflect minimum and maximum number of entries, plot sizes, seed quantity, and yield measurement strategies—tailored to crop and trial purpose.
2. Ensure all measured traits in tricot trials are directly linked to TPPs to maintain relevance for product advancement and seed system targeting.
3. Include mandatory metadata fields (e.g., trial purpose, candidate vs. check variety) and develop a shared glossary to improve consistency and clarity across teams.
4. Improve trial documentation to explicitly state objectives (e.g., verification, seed system testing) and ensure alignment with breeding and scaling strategies.
5. Create guidelines that allow variation in entries based on trial type (pipeline, released material, market research), while maintaining alignment with breeding goals and TPP thresholds.
6. Conduct pre-season training for data collectors and field agents to ensure standardized implementation of protocols and accurate data collection.
7. Engage with national variety release authorities to align tricot yield measurements with their data requirements and gain regulatory acceptance.
8. Integrate tools in ClimMob to support yield validation, trial classification, and statistical methods to account for seed loss and trial attrition.
9. Implement a system of default check varieties by crop and market segment to enable cross-trial analysis and maintain internal consistency.
10. Shift the narrative to position tricot as a flexible, decision-support platform that strengthens data collection for variety selection and market integration.



Joost van Heerwaarden (Wageningen University and Research) presenting the framework to estimate genetic gain on-farm using data generated in tricot trials.

Data analysis and decision-making

The third session of the meeting focused on one of the most critical stages of the tricot process, data analysis and use to inform robust, transparent, and scalable decision-making. The discussions across all three groups emphasized the need for enhanced reporting tools, structured review processes, and sustained capacity development to fully leverage the potential of tricot trials.

A key theme was how to present OFVT data in a way that supports actionable decisions by breeding teams, product development teams (PDTs), and other stakeholders. Participants agreed that reports must go beyond simple trait comparisons. They should include yield advantage estimates, spatial distribution of performance, comparisons with checks and popular varieties, and indicators such as willingness to replant. The reports must be modular, customizable, and tailored to diverse users, from data scientists to farmers and national authorities. Tools such as histograms, bi-plots, segmented analyses, and combined year-location summaries were suggested to make the insights both rigorous and accessible.

In ensuring this data feeds into real-world decisions, the role of PDTs in advancement decisions was heavily emphasized. All groups recommended establishing routine, structured advancement meetings with multidisciplinary PDTs, ideally integrated into an annual calendar and segmented by market or pipeline stage. Participants stressed the importance of defining minimum data thresholds and ensuring timely availability of both on-farm and on-station data. Where possible, pre-meeting data summaries and cross-country analyses should be shared to support consistent, well-informed discussions. Documentation of the decisions (including rationale, expectations for the next cycle, and follow-up questions) was seen as vital for transparency and continuous learning.

The session also tackled how OFVT data can support the refinement of Market Segments (MS) and TPPs**.** Participants proposed reviewing adoption and market feedback data every 3–5 years and incorporating findings from farmer preferences and socio-economic segmentation into updated profiles. There was a clear call for more frequent market research (e.g., every 2–3 years) and for expanding the diversity within trial designs to test multiple segments in a single setup. A suggested approach is the research developed by the Market Intelligence Initiative team[2] on video-based product concept testing with farmers. Recognizing that MS and TPPs are dynamic rather than fixed, the session advocated for systematic analysis of trait drivers and updating the weightings used in selection indices accordingly.

On the topic of routine data sharing and utilization, all groups envisioned a structured process starting from pre-season planning and culminating in post-meeting documentation. Full calendars of pre-planning, trialing, analysis, and reporting cycles were recommended. These would be complemented by regular (e.g., bi-monthly or quarterly) PDT update meetings, coordination with other decision-making groups, and communication platforms that ensure information reaches breeders, field teams, and decision-makers in a timely manner. Importantly, participants also emphasized providing feedback to farmers before the next planting season to build trust and close the loop between research and practice.

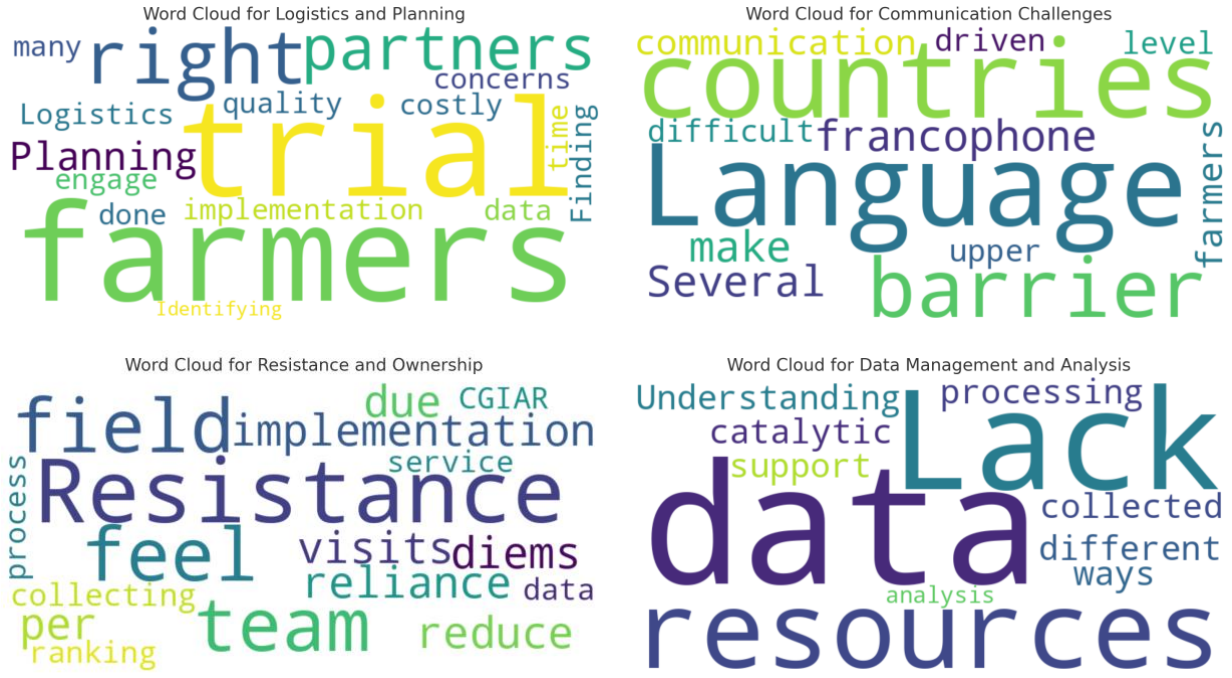
[2]Donovan et al. (2025) Experimental Agriculture vol 61, e5, 1-18. https://doi.org/10.1017/S001447972500002X

To enable all these improvements, the groups agreed that capacity development is crucial**.** Training is needed across the board: PDT members must be able to interpret data and use tricot tools; field agents must understand protocols and communicate effectively with farmers; and data scientists need familiarity with breeding platforms like ClimMob. Suggestions included developing case studies, simplified toolkits, and shared learning platforms to support ongoing capacity-building efforts. Finally, stronger communication infrastructure (including feedback mechanisms and shared dashboards) was recommended to reinforce learning and data-driven culture.

In summary, this session laid out a clear vision for how tricot can move from data collection to strategic influence. With better tools, structured processes, and stronger capacities, tricot has the potential to become a central mechanism for guiding breeding decisions, refining market strategies, and ensuring that improved varieties meet the real needs of farmers.

KEY TAKEAWAYS FROM SESSION 3: DATA ANALYSIS AND DECISION-MAKING

1. Design customizable reports that present key metrics like yield advantage, spatial distribution, and trait performance tailored for PDTs, breeders, and policy audiences.
2. Define and document clear thresholds and trait metrics (e.g., yield, preference, willingness to replant) in SOPs to support advancement decisions using OFVT data.
3. Create a country-level, market-segment-specific calendar for pre-season planning, data analysis, and structured PDT advancement meetings.
4. Develop or integrate tools (e.g., Advancement Decision Capture Tool) to record the logic behind decisions and expectations for the next cycle.
5. Ensure that all trial entries and data collection protocols are explicitly aligned with well-defined market segments and target product profiles.
6. Conduct adoption reviews and market research every 2–3 years, and use results to adjust product profiles, selection indices, and trait priorities.
7. Create mechanisms for systematic sharing of on-station, lab, and on-farm data across breeding pipelines, from early-stage trials to OFVTs.
8. Develop targeted training programs, toolkits, and reflection platforms for PDT members, field agents, and data scientists.
9. Integrate tools within ClimMob for multi-year, multi-location, and segmented data analysis, with visualization options (e.g., histograms, bi-plots, preference drivers).
10. Establish systems to communicate key insights and decisions to farmers before the next season to maintain trust and participation.



Word clouds with the most common terms used by implementing teams in the CGIAR-NARES Network when reporting the main challenges when operationalize and coordinate tricot trials at scale. In clockwise direction from top left: challenges in logistics and planning; challenges in communication; challenges in data management and analysis; challenges in NARES feeling the ownership of the approach.

Operationalization, coordination and scaling

The fourth session of the meeting turned its focus toward the operational aspects of implementing tricot trials at scale, addressing practical challenges around cost, coordination, logistics, and long-term sustainability. Across all breakout groups, participants emphasized that while the tricot approach is powerful and flexible, its success depends heavily on how well it is planned, resourced, and managed across diverse country contexts.

One of the primary concerns raised was the cost of implementing tricot trials. The most significant drivers included seed multiplication and distribution, payments to field agents and extension officers, logistics (especially transportation and fuel), data collection tools (e.g., booklets or digital devices), packaging materials, and general coordination costs. Additionally, training and capacity building for farmers, partners, and implementing staff represented a major investment, but one considered essential for quality and scale.

To improve the efficiency and cost-effectiveness of OFVT using tricot, participants recommended several approaches. These included aligning trials more closely with breeding pipelines, limiting traits to those most essential for decision-making, and optimizing sample sizes. The frequency of field visits could also be reduced to only critical moments such as planting and harvest. Some teams proposed leveraging lead farmers or centralized data submission points (e.g., through local chiefs or farmer groups) to reduce the burden on field agents. Others suggested building a sentinel network of trained farmers (core group of participants who could be relied on for consistent data collection), thus reducing dependence on extension systems over time.

Several groups highlighted the importance of early, detailed, and structured planning. Ideally, preparations should begin 6 to 12 months before trials are implemented. This planning window allows for seed multiplication, procurement of materials, stakeholder engagement, and the allocation of human and financial resources. Some teams recommended creating seasonal or annual planning calendars, complete with checklists and SOPs, to streamline operations and prevent last-minute challenges. Integration of trial planning into the post-advancement process was seen as a key opportunity for proactive coordination.

When discussing multi-crop coordination at the country level, participants envisioned networks organized around agroecological zones, cropping calendars, and local administrative units. These networks would allow shared planning, pooled resources, and stronger collaboration between breeding programs, extension services, cooperatives, NGOs, and research institutions. Examples were drawn from Rwanda and Malawi, where organizations have begun experimenting with such models through coordinated efforts led by CGIAR centers and national partners.

Beyond technical and logistical considerations, participants underscored the need for motivating stakeholders. Both monetary incentives (such as stipends or transport reimbursement) and non-monetary incentives (like branded merchandise, training opportunities, or recognition) were considered useful. The idea of establishing catalytic funds (pooled resources used to support coordinated, multi-crop trials) was raised to reduce fragmentation and promote efficiency.

Finally, the discussion called for finalizing harmonized SOPs and operational guidelines, improving communication across implementing partners, and building a culture of

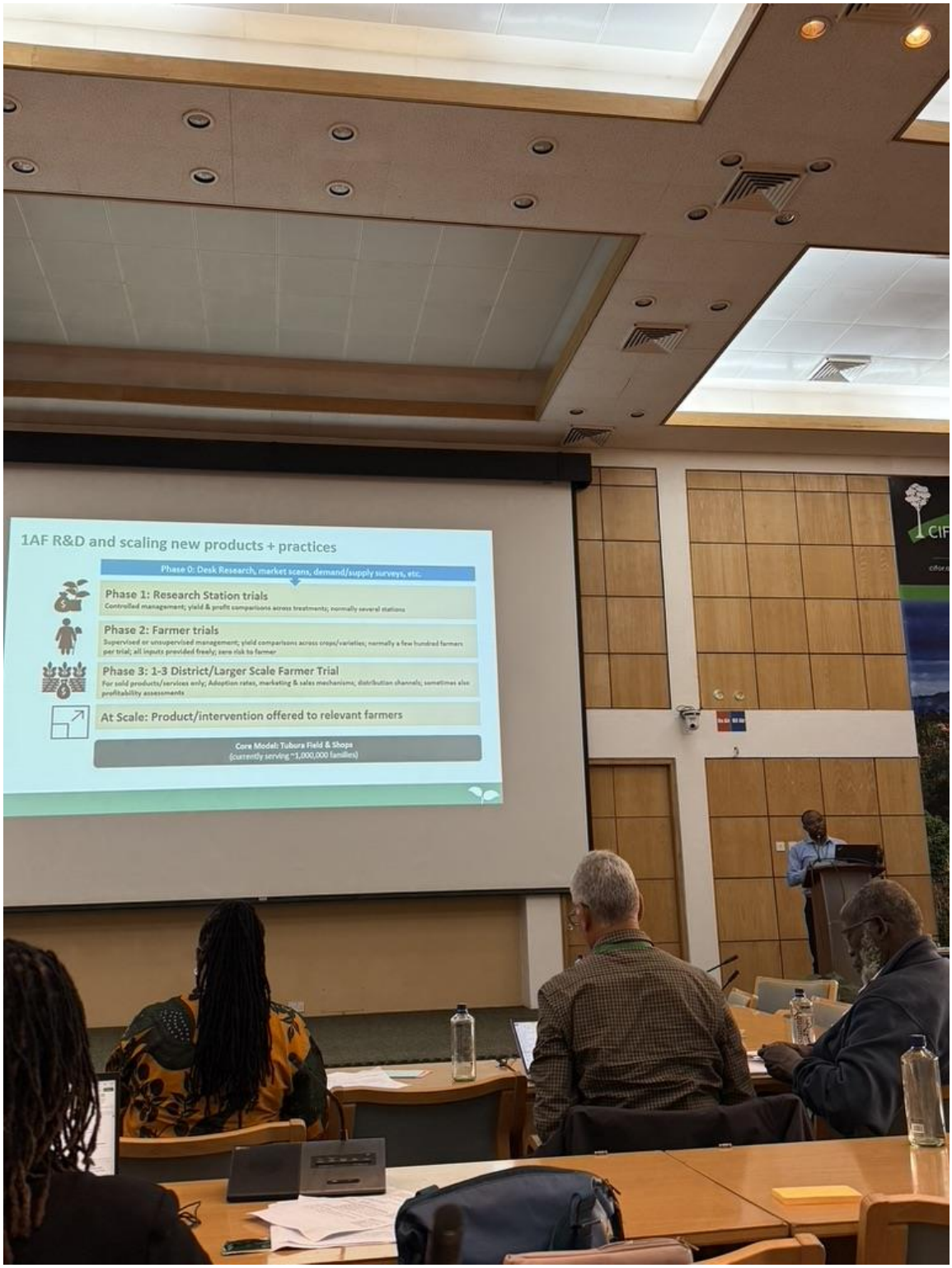
accountability and learning. While each region and crop may require specific adaptations, common tools for planning, budgeting, and coordination were seen as essential to scale up the tricot model without compromising data quality or farmer experience.

This session made it clear that tricot's continued expansion depends not only on scientific rigor, but also on strong partnerships, efficient systems, and thoughtful operational planning.

KEY TAKEAWAYS FROM SESSION 5: OPERATIONALIZATION, COORDINATION AND SCALING

1. Trial implementers should begin planning immediately after the advancement meeting, aligning with cropping seasons and ensuring timely seed multiplication, logistics, and partner engagement.
2. Identify key cost components (e.g., seed, transport, data tools, agent payments) early and apply strategies such as reducing traits, optimizing sample size, and clustering trials geographically.
3. Build a network of skilled and motivated farmers who can consistently participate in trials and collect data, reducing long-term reliance on extension agents.
4. Establish community-level data submission points (e.g., through lead farmers or local coordinators) to lower logistical and labor costs.
5. Finalize SOPs that define logistics, roles, and cost norms across crops and regions. Include standard tools for costing, sample size calculation, and planning timelines.
6. Facilitate coordination across breeding programs, NARES, NGOs, and cooperatives by creating national-level teams or platforms to plan and implement multi-crop trials.
7. Pool resources from multiple programs or partners to fund coordinated trials at scale, using shared infrastructure and minimizing duplication.
8. Clearly define who is responsible for each aspect of trial operations (such as seed logistics, data management, training, and farmer coordination) across CGIAR centers, NARES, and local partners to improve accountability and reduce delays.
9. Train field agents, coordinators, NARES partners, and farmer-facing organizations on tricot methodology, soft skills, and operational planning.
10. Limit field visits to critical stages like planting and harvest, and conduct random data quality audits on a sample of trial sites to ensure reliability at scale.





Elysée Tuyishime, shares the experience of One Acre Fund in implementing tricot as part of the R&D routines in Rwanda.

Action points from cross-cutting teams

The final session of the meeting brought participants together to reflect on the cross-cutting themes that underpin successful implementation of the tricot approach. Building on the insights and challenges raised during the earlier sessions, the goal was to generate concrete, forward-looking action plans that will improve coordination, decision-making, digital support, and standardization across all tricot trials. Participants divided into four breakout groups focused on key operational pillars: the advancement process, cross-crop coordination networks, the digital ecosystem, and harmonized standard operating procedures (SOPs).

Each group worked toward identifying practical steps for the next 12 months, grounded in shared principles and aligned with the broader vision of scalable, efficient, and inclusive on-farm validation trials. Discussions emphasized the importance of building institutional ownership, strengthening interoperability between tools, aligning data and decision processes with market segments and TPPs, and ensuring that protocols are both rigorous and user-friendly. The resulting action plans offer a roadmap for translating strategic intent into coordinated action, helping tricot evolve from a method into a system that supports the future of OFVT and scaling at CGIAR-NARES.

Advancement Process

The breakout group on the Advancement Process focused on refining decision-making protocols for OFVTs using the tricot approach, ensuring decisions are data-driven, inclusive, and aligned with market segment (MS) and target product profiles (TPPs).

Key principles

- Inclusivity: Ensure diversity in farmer demographics and agro-ecologies.
- Data-driven decisions: Base advancement on quality, multi-year data, including on-farm, on-station, and lab sources.
- Pre-agreed criteria: Use MS/TPPs and clear thresholds to guide variety advancement at each stage.
- Stakeholder engagement: Include the right people (e.g., PDT members, NARES, breeders, extension agents) at the right decision points.
- Scalability and multidisciplinarity: Build systems that work across regions, crops, and disciplines.
- Continuous Improvement: Use each season's learning to improve the next.

Recommendations and priorities for future investment

- Clearly define the role and weight of different data types in decision-making.
- Improve planning and pre-trial engagement, especially with Product Design Teams (PDTs).
- Build capacity in interpreting data, conducting trials, and facilitating the advancement process across different actors.



- Develop an Advancement Process Toolkit to support country-level teams, along with reporting templates and decision logs.

Action plan for the next 12 months

- Clarify responsibilities at each stage of the advancement process (ABI level, crop level, country level).
- Define what data is needed at different advancement stages and ensure alignment with country-level Target Product Profiles (TPPs).
- Outline and formalize the full advancement process steps, including: a) Annual planning of trials at country levels – late stage, OFVT; b) Pre-season planning; c) Conduct of trials, it's monitoring; d) Data collection; e) Data analysis; f) Review of data and analysis by Internal technical teams; g) Review of data and analysis with PDT by market segment.
- Develop crop- and country-specific models for the advancement process.
- Implement the full process across selected crops in its first year.
- Establish advancement teams for each crop and country.
- Identify and address capacity-sharing needs for key actors, including breeders, extension agents, and farmers.
- Create standardized reporting templates for advancement meetings.
- Promote cross-crop learning on how the advancement process is conducted.

Cross-crop Coordination Network

The breakout group for the Cross-crop Coordination Network focused on how to strengthen coordination across crops and countries to improve efficiency, reduce duplication, and enhance learning in tricot OFVT.

Key principles

- Structured coordination: Establish coordination units at country, regional, and global levels with clear roles and responsibilities, ideally backed by dedicated budgets.
- Institutionalization: Embed coordination within national research systems using a RACI framework (Responsible, Accountable, Consulted, Informed) to ensure ownership and accountability.
- Scalability: Ensure mechanisms allow scaling of both practices and coordination efforts.
- Separation of functions: Recognize different coordination needs pre- and post-variety release.
- Clarity of purpose: Clearly define trial objectives (e.g., verification) and ensure transparency and documentation.
- Cost-effectiveness and cost-sharing: Enable shared funding mechanisms or pooled resources to make coordination financially viable.
- Communication and leadership: Prioritize effective communication and active engagement with institutional leaders.

Recommendations and priorities for future investment

- Conduct a mapping of existing cross-crop coordination networks to understand current structures and gaps.
- Develop Standard Operating Procedures (SOPs) to guide how these networks function.
- Mobilize resources for coordination, including integrating fundraising efforts with breeding teams.
- Promote training and awareness at institutional levels to foster ownership of tricot-based OFVT.

Action plan for the next 12 months

- Map existing cross-crop coordination networks in at least three countries per region.
- Mobilize resources in collaboration with breeding teams.
- Conduct institutional training and sensitization to embed tricot in national systems.

Digital Ecosystem

The Digital Ecosystem group focused on advancing the digital infrastructure supporting tricot trials by promoting interoperability, governance, and user experience across key tools and investments like ClimMob, EBS, BioFlow, AgPile, and others.

Key principles

- Interconnected ecosystem: Systems must function together seamlessly while retaining independent utility. This includes ClimMob, AgPile, Breeding Portal, BioFlow, and phenotyping tools (e.g., Artemis).
- Governance first: Define data distribution rules and metadata standards up front. Governance should shape the technical design, ensuring FAIR principles (Findable, Accessible, Interoperable, Reusable)—without assuming open access at principle.
- Data architecture & refactoring: A flexible data warehouse architecture is needed, with close collaboration between 1000FARMS and AgPile teams.
- Stakeholder-centered: Understanding user needs is essential. Stakeholders include breeders, trial managers, and data scientists. A product manager role is recommended.
- Hybrid data collection: Multiple modes of data entry (ODK, USSD, IVR, QED, paper) should be supported to accommodate different contexts.
- Future-oriented integration: Include innovations such as computer vision, audio tools, and AI from related projects.

Recommendations and priorities for future investment

- Set governance rules that reflect ongoing operations.
- Collaborate with digital platforms like AgPile for back-end refactoring and analytics workflows.
- Define metadata standards and identify shared datasets by priority use cases.



- Improve ClimMob functionality:
    - Backend refactoring into a data warehouse.
    - Improved UX/UI.
    - Better documentation and helpdesk support.
    - Time-based reminders and flexible data registration.
    - SOP implementation aligned with crop ontologies.

Action plan for the next 12 months

- Design and initiate a data warehouse system to unify digital tools.
- Integrate ClimMob with BioFlow to improve workflow interoperability.
- Improve product documentation and user testing materials.
- Organize a support team with clear roles and issue-response structures.
- Redesign ClimMob's project creation interface, aligning it with crop ontologies SOPs and crop networks.
- Contribute to the CGIAR-wide governance conversation on digital tools and data.
- Share analytical workflows and tricot data use cases across CGIAR and partners.

Standard Operating Procedures (SOPs)

This group focused on proposing a unified, purpose-led framework for implementing tricot OFVT through clear and practical SOPs, ensuring consistency, quality, and institutional adoption.

Key Principles

- Purpose-driven SOPs: SOPs should reflect the specific purpose of each OFVT (e.g., product advancement, variety release, adaptation testing).
- Full-cycle process mapping: SOPs should cover all stages—from planning and trial design to data management, analysis, and decision-making.
- Inclusivity & Feedback: SOP development and validation must involve implementing partners, end-users, and variety release committees.
- Institutionalization: SOPs should be formalized within breeding programs, with clear commitments to their use and regular review to allow innovation.
- Innovative approaches: Encourage experimentation and field testing of new methods within OFVT frameworks.

Recommendations and priorities for future investment

- Identify relevant partners and define clear roles for SOP development.
- Establish a committee to oversee SOP processes, backed with dedicated funding.
- Make the CGIAR Accelerated Breeding Initiative (ABI) responsible for SOP development, with crop teams' support.
- Create a Community of Practice (CoP) to collaboratively develop, review, and update SOPs (both cross-cutting and crop-specific).
- Develop decision-making models to guide testing strategies and manage trade-offs.
- Update SOPs to reflect evolving data needs, such as number of traits collected.

- Promote capacity sharing and broader stakeholder engagement on SOP implementation.
- SOPs should be linked to crop ontologies for seamless repurposing and integration with other databases.

Action plan for the next 12 months

- Secure financing for process mapping and SOP development.
- Initiate process mapping and set up the CoP across relevant teams.
- Develop a prototype of the decision-making model to guide trial design and prioritization.
- Co-develop SOPs linked to crop ontologies with ABI, crop teams, and the 1000FARMS team.
- Prepare and circulate OFVT process guidelines for stakeholder use.

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